



Muhammad Harish

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ABOUT ME

Robotics research engineer with a strong foundation in control systems, embedded design, and artificial intelligence. Experienced in developing autonomous robots integrating sensors (IMU, LiDAR, cameras), embedded systems (Jetson, Raspberry, Arduino, Espressif), and high-level perception algorithms using ROS, Python, and C++. Passionate about bridging hardware and intelligence through research in Robotics and AI.

RESEARCH INTERESTS

Autonomous robot navigation, SLAM and visual-inertial odometry (VIO), learning-based control, and robotic perception.

WORK EXPERIENCE

🏢 **Lab Inovasi Nusantara** – Jakarta, Indonesia

Robotics Developer

[Jun 2025 – Current]

Designed and implemented an autonomous aerial inspection drone equipped with Jetson Xavier, Pixhawk Cube, and Livox Mid-360 LiDAR for 3D environmental mapping. Implemented FAST-LIO2 SLAM, VIO, and EGO-Planner path planning.

🏢 **PT Prima Medika Laboratories (Pharos Group)** – Jakarta, Indonesia

Robotics Engineer

[Sep 2024 – Current]

Conducted research and development on robotic and autonomous systems, focusing on perception, control, and embedded intelligence. Designed a semi-autonomous ASRS robot with ROS-based obstacle detection and path planning.

EDUCATION AND TRAINING

Bachelor of Electrical Engineering

Universitas Pertamina [Sep 2021 – May 2025]

City: Jakarta | Country: Indonesia | Website: <https://universitaspertamina.ac.id/> | Final grade: GPA: 3.96/4.00 | Thesis: Integration of Kinematics and Reinforcement Learning for Hexapod Locomotion

Control Systems, Robotics, Artificial Intelligence, and Automation

PUBLICATIONS

[2025]

Implementation of Reinforcement Learning on a Hexapod Robot Undergraduate Final Thesis Paper

Authors: Muhammad Harish | Publisher: Repository Pertamina University

PROJECTS

[Jun 2025 – Current]

SKYLINE- Autonomous Inspection Drone Designed and implemented an autonomous aerial inspection drone equipped with Jetson Xavier, Pixhawk Cube, and Livox Mid-360 LiDAR. Implemented FAST-LIO2 SLAM, VIO, and EGO-Planner in ROS (Linux) for real-time localization, mapping, and path planning. Focused on sensor fusion and autonomous flight algorithms for aerial robotics applications.

[Feb 2023 – Sep 2024]

SKYRONE: Autonomous Multirotor Robot for Search and Rescue Led the design, integration, and testing of a ROS-based autonomous VTOL drone with computer vision and flight control systems for the KRTI (National aerial robotics competition)

Link: <https://www.youtube.com/watch?v=KWGLskeG1Dg>

[Dec 2022 – Aug 2024]

RETRO-BOT: Hexapod Robot for Disaster Search and Rescue Led the development of a six-legged autonomous rescue robot integrating robotics and AI technologies. Implemented computer vision and machine learning for victim detection, with inverse kinematics and adaptive gait algorithms using reinforcement learning for stable locomotion. Designed for the KRI (National robotics competition)

Link: <https://www.youtube.com/watch?v=0PW9O3eiulK>

HONOURS AND AWARDS

[Dec 2025] Indonesian Academy of Young Scientists - ALMI

1st Place - National Thesis Award Awarded for an outstanding undergraduate thesis on integrating kinematic constraints with reinforcement learning for stable and adaptive hexapod robot locomotion in complex environments.

[Feb 2024] Universitas Pertamina

1st Place- The Most Outstanding Student Recognized as the top student among 15 faculties for academic excellence (GPA 3.96/4.00), leadership, and contribution to research and innovation.

[Sep 2023] Ministry of Education and Research, Indonesia

1st Place – National IT and Innovation Competition (GEMASTIK) Led the team to national victory with *RoadSense*, a deep learning based road damage detection system using optimized YOLOv5 and depth camera using intel realsense for real-time infrastructure analysis.

[Mar 2023] Institut Teknologi Sepuluh Nopember

1st Place – National Paper Competition (BARONAS) Developed *ROAD-MAP*, an IoT and edge computing-based road condition mapping system integrating deep learning for damage assessment.

[Oct 2022] Universitas Airlangga

1st Place – Airlangga Innovation Competition Created *Trafo*, an AI-driven trash classification system using computer vision for sustainable waste management.

[2024] Ministry of Law and Human Rights, Indonesia

Registered Intellectual Properties Five granted innovations: *RoadSense*, *IRIS (IoT Shrimp Counter)*, *EMA (Smart Energy Monitor)*, *MA P-Road*, and *WeCan-See (AI Visual Assistance)* — covering IoT, AI, and robotic sensing applications.

NETWORKS AND MEMBERSHIPS

[Dec 2022 – Aug 2024] Universitas Pertamina

Director - Robotics Student Activity Unit Overseeing a team of 17 staff and 60 members to execute 15 work programs, including Company Visits, Research, Career Information, Software Training, Robotics Competitions, Workshops, and Knowledge Sharing sessions. Oversaw and coached UAV and legged robot teams in engineering, project management, and innovation.

Link: <https://www.instagram.com/retro.uper/>

CERTIFICATIONS

[Universitas Pertamina, Feb 2025]

TOEFL ITP Score: 583

[Udemy, 2025]

Udemy Course ROS for Beginners I – III: Basics, Motion, OpenCV, Localization, Navigation, SLAM, Web-based Navigation with ROSBridge. PLC Programming and Hardware Level I – III: PLC Fundamentals, Applied Logic, Process Visualization. Modern Computer Vision GPT, PyTorch, Keras, OpenCV4 in 2024.